API Integration (Weather App using Tkinter GUI)

Objective

To develop a Python GUI application that fetches live weather information from the OpenWeatherMap API using the requests module and displays the parsed JSON response.

Requirements Covered

• Use requests module
• Parse JSON response
• Add search/filter (search by city name)
• Use Tkinter GUI
• Exception handling with try-except
• Python script + GitHub link
• Screenshots of output

Software Requirements

• Python 3.x
• requests library
• Tkinter (built-in with Python)
• Internet connection

API Used

OpenWeatherMap Current Weather API
https://openweathermap.org/api

Project Description

The application allows the user to enter a city name and retrieve live weather data such as temperature, humidity, pressure, weather condition, and wind speed.

Algorithm

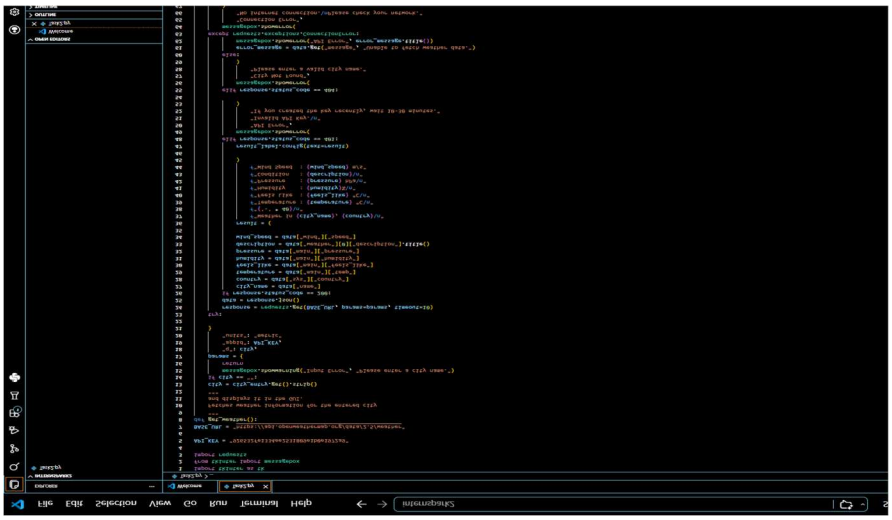
1. Accept city name from the user.
2. Validate the input.
3. Send request to OpenWeatherMap API.
4. Receive JSON response.
5. Parse required weather fields.
6. Display the information in the GUI.

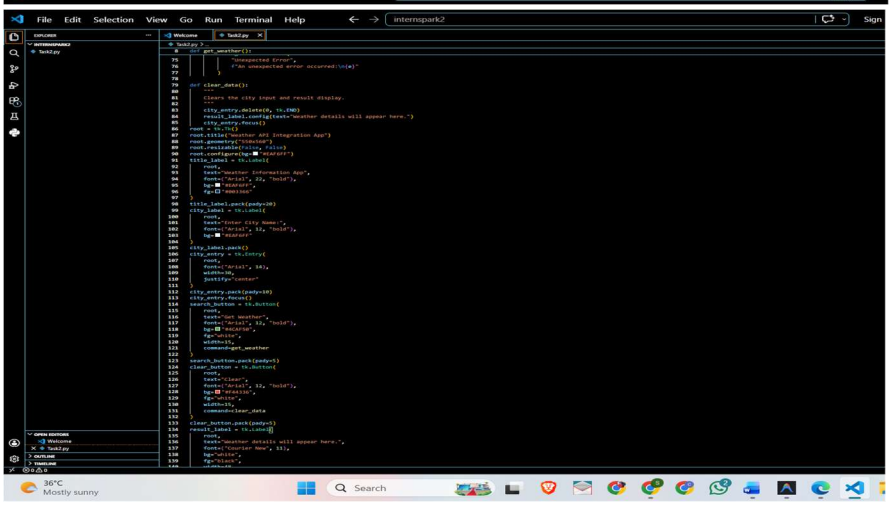
7. Handle errors.
8. Provide a Clear button.

Installation Steps

1. pip install requests
2. Save the file as weather_api_gui.py
3. Replace API_KEY = "YOUR_API_KEY_HERE" with your actual key.
4. Run: python weather_api_gui.py

Source Code

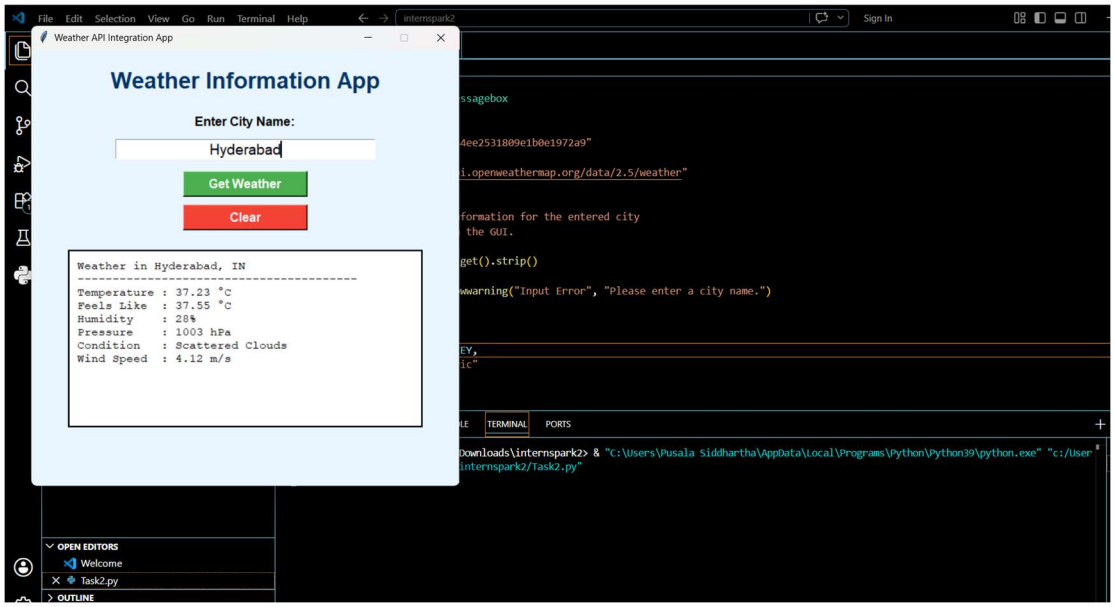




Sample Input

Enter City Name: Hyderabad

Input and Output Screenshot



GitHub Repository Link

https://github.com/Siddharthapusala/API-Integration-Weather-App-.git

Conclusion

This project successfully demonstrates API integration using requests, JSON parsing, search functionality, and Tkinter GUI development. All internship requirements are satisfied.